

Time Series Analysis of Ventura County Wildfires

Using SARIMA and ARMAX Models

Edgar Gonzalez Martinez - gonzalezmartinez@ucsb.edu

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This project studies monthly wildfire activity in Ventura County, California from February 2000 to December 2020. The wildfire data were aggregated into monthly fire counts and total acres burned. First, I used a SARIMA model to analyze monthly wildfire starts using the transformed series $\log(1 + x)$. The selected SARIMA model was $ARIMA(0, 1, 3)(0, 0, 1)_{12}$, and the Ljung-Box test suggested that the residuals did not have strong remaining autocorrelation.

For the second part of the project, I modeled monthly acres burned using an ARMAX model with external predictors. I tested drought conditions from SPI, weather variables from NASA POWER, vegetation greenness using NDVI, and a simplified Santa Ana wind event indicator. BIC selection chose lagged severe drought, Santa Ana-like wind events, and mean NDVI as the final predictors for acres burned. The ARMAX model improved AICc and BIC compared with the baseline model and had acceptable residual diagnostics. Overall, the results suggest that wildfire severity in Ventura County is connected to drought, vegetation conditions, and Santa Ana-like wind patterns.

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1 Introduction

Wildfires are an important issue in California because they can affect land, communities, air quality, and public safety. In this project, I study monthly wildfire activity in Ventura County, California from February 2000 to December 2020. I chose Ventura County because it is my home county, so the topic is personally interesting to me. The dataset also works well for time series analysis because the fire records can be aggregated by month into wildfire starts and total acres burned.

The purpose of this project is to model wildfire activity using methods from time series analysis. I first use a SARIMA model to study monthly wildfire starts. This gives a baseline model that only uses the past behavior of the fire count series. Then, I use an ARMAX model to study monthly acres burned, which represents wildfire severity. For the ARMAX model, I include external predictors related to drought, weather, Santa Ana-like wind events, and vegetation conditions.

Past wildfire studies often focus on how climate and environmental conditions are connected to fire activity. In this project, I use that same general idea by testing whether drought, wind, humidity, temperature, vapor pressure deficit, and NDVI help explain monthly acres burned. The main discovery is that the final ARMAX model selected lagged severe drought, Santa Ana-like wind conditions, and mean NDVI as useful predictors. This suggests that wildfire severity in Ventura County is not only related to past fire behavior, but also to drought, vegetation, and Southern California wind patterns.

2 Data

The main wildfire dataset used in this project comes from the USDA Forest Service Fire Program Analysis Fire-Occurrence Database. The original records include information such as discovery date, final fire size, and fire location. I filtered the data to Ventura County, California and aggregated the individual wildfire records into a monthly time series.

The original wildfire file covers January 1992 through December 2020. However, for the final analysis, I restrict the sample to February 2000 through December 2020. I use this common time period because the NDVI vegetation data begin around 2000, and using the same general period makes the SARIMA and ARMAX analyses more comparable. The final monthly wildfire sample contains 251 monthly observations.

The two main wildfire variables are monthly fire count and monthly total acres burned. The fire count variable measures the number of wildfire starts in Ventura County each month. In the 2000-2020 sample, monthly fire count ranges from 0 to 19, with a median of 1 and a mean of about 2.23. This shows that many months had low fire activity, while a few months had much higher wildfire counts.

The total acres burned variable measures the total fire size for each month. This variable is much more extreme than fire count because one large wildfire can dominate the total acres burned for an entire month. In the 2000-2020 sample, monthly total acres burned ranges from 0 to about 281,899.5 acres, with a median of about 0.10 acres. This shows that most months had very low acres burned, but a few extreme wildfire months had very large values.

For the SARIMA portion of the project, I use monthly fire count from February 2000 through December 2020. Since the fire count series contains many zero values and some spikes, I apply the

transformation $\log(1 + \text{fire count})$. The $+1$ keeps zero-fire months in the analysis, while the log transformation reduces the effect of unusually large monthly fire counts.

For the ARMAX portion of the project, I use monthly acres burned as the response variable and add external drought, weather, wind, and vegetation predictors. Acres burned is transformed using $\log(1 + \text{total acres burned})$ because the original acres-burned series is highly right-skewed and contains several extreme wildfire months.

The drought variables come from the SPI drought data file. I use the drought categories D0 through D4, where larger categories represent more severe dry conditions. I also create lagged drought variables at 1-month, 3-month, and 6-month lags because drought may affect wildfire severity after some delay. In addition, I create a weighted drought index using $D0 + 2D1 + 3D2 + 4D3 + 5D4$ so that more severe drought categories receive more weight.

Weather variables are taken from NASA POWER daily data and then aggregated to the monthly level. These variables include maximum temperature, wind speed, relative humidity, and vapor pressure deficit. I also create a simplified Santa Ana-like wind variable using wind direction, maximum wind speed, and relative humidity. This variable counts dry offshore-wind days in each month and is included because Santa Ana winds are an important fire-weather pattern in Southern California.

Vegetation conditions are measured using NDVI from MODIS. NDVI is used as a measure of vegetation greenness and possible fuel condition. After merging the wildfire, SPI, weather, Santa Ana wind, and NDVI variables, the final ARMAX dataset contains 248 complete monthly observations. This smaller number occurs because lagged drought variables require earlier months to be removed from the final modeling dataset.

3 Methodology

This project uses two main time series methods. First, I use a SARIMA model to analyze monthly wildfire starts in Ventura County. Second, I use an ARMAX model to analyze monthly acres burned while including external predictors related to drought, weather, Santa Ana-like wind events, and vegetation conditions. Both parts of the final analysis use the common sample from February 2000 through December 2020.

3.1 SARIMA Model

For the first part of the analysis, I model monthly wildfire starts using a SARIMA model. The response variable is monthly fire count. Since the fire count series contains zero months and several spikes, I use the transformation

$$y_t = \log(1 + x_t)$$

where x_t is the monthly number of wildfire starts. The value 1 is added so that months with zero wildfire starts can still be included, since $\log(0)$ is undefined.

Because the data are monthly, I define the time series with frequency 12. This allows the model to account for possible yearly seasonal patterns. I use the Box-Jenkins approach by examining

the time series plot, ACF plot, and PACF plot of the transformed series. I also use `ndiffs()` and `nsdiffs()` to help determine whether regular or seasonal differencing is needed.

The general SARIMA model has the form

$$SARIMA(p, d, q)(P, D, Q)_{12}$$

where p, d, q are the nonseasonal ARIMA orders, P, D, Q are the seasonal ARIMA orders, and 12 is the seasonal period for monthly data. I compare candidate models using AICc and BIC. I also use `auto.arima()` with `seasonal = TRUE` to help select the final model automatically.

After selecting the SARIMA model, I check the residuals using residual plots, the residual ACF, and the Ljung-Box test. The goal is for the residuals to look approximately like white noise. A Ljung-Box p-value above 0.05 suggests that there is not strong evidence of remaining autocorrelation.

3.2 ARMAX Model

For the second part of the analysis, I model monthly acres burned using an ARMAX model. Acres burned is used as the wildfire severity response. Since acres burned is highly right-skewed and contains several extreme wildfire months, I use the transformation

$$z_t = \log(1 + \text{total acres burned}_t)$$

This transformation reduces the effect of extremely large wildfire months while still keeping months with zero acres burned in the analysis.

The ARMAX model combines regression predictors with an ARIMA error process. The regression part can be written as

$$z_t = \beta_0 + \beta_1 X_{1,t} + \beta_2 X_{2,t} + \dots + \beta_k X_{k,t} + e_t$$

where z_t is the transformed acres-burned response, $X_{1,t}, X_{2,t}, \dots, X_{k,t}$ are external predictors, and e_t follows an ARIMA process. This allows the model to include drought, weather, wind, and vegetation variables while also accounting for time dependence in the residuals.

The external predictors include SPI drought categories, NASA POWER weather variables, a simplified Santa Ana-like wind indicator, and MODIS NDVI. The SPI drought categories D0 through D4 are lagged by 1, 3, and 6 months because drought may affect wildfire severity after some delay. I also create a weighted drought index:

$$\text{Weighted Drought Index} = D0 + 2D1 + 3D2 + 4D3 + 5D4$$

This gives more weight to more severe drought categories.

Vapor pressure deficit is included as a measure of atmospheric dryness. I calculate saturation vapor pressure as

$$e_s = 0.6108 \exp\left(\frac{17.27T}{T + 237.3}\right)$$

and vapor pressure deficit as

$$VPD = e_s \left(1 - \frac{RH}{100}\right)$$

where T is mean temperature and RH is relative humidity.

I also create a simplified Santa Ana-like wind event variable using daily wind direction, maximum wind speed, and relative humidity. A day is counted as Santa Ana-like if it has northeast/east-sector wind direction, moderate-to-strong wind, and low relative humidity. These daily indicators are then aggregated to the monthly level as `santa_ana_days` and `santa_ana_flag`.

To choose the final ARMAX predictors, I first fit a large regression model using the possible drought, weather, Santa Ana wind, VPD, and NDVI variables. I then use BIC stepwise selection with

$$k = \log(n)$$

so that the selection penalizes unnecessary predictors more strongly than AIC. After BIC selects the predictors, I check VIF values to make sure multicollinearity is not a major issue.

Finally, I fit the ARMAX model using the BIC-selected predictors and compare it to a baseline SARIMA model for acres burned on the same 2000-2020 sample. The comparison is based on AICc, BIC, and residual diagnostics. I also use the Ljung-Box test to check whether the ARMAX residuals still show significant autocorrelation.

4 Results

4.1 SARIMA Results for Monthly Wildfire Starts

The first part of the analysis modeled monthly wildfire starts in Ventura County using the transformed fire count series from February 2000 through December 2020. After applying the transformation

$$y_t = \log(1 + \text{fire count}_t),$$

I examined the time series plot, ACF plot, and PACF plot. The ACF showed slow decay, suggesting that regular differencing was needed. This matched the output from `ndiffs()`, which suggested one regular difference. The output from `nsdiffs()` suggested that seasonal differencing was not needed.

The model selected by `auto.arima()` was

$$ARIMA(0, 1, 3)(0, 0, 1)_{12}.$$

This model uses one regular difference, three nonseasonal moving-average terms, and one seasonal moving-average term with period 12. The selected SARIMA model had AICc about 467.92 and BIC about 485.28.

The residual diagnostics were acceptable. The Ljung-Box test gave a p-value of about 0.1224, which is above 0.05. Therefore, I do not reject the null hypothesis that the residuals are white noise. This suggests that the SARIMA model captured most of the autocorrelation in the transformed monthly wildfire count series.

The 12-month forecast from the SARIMA model predicted monthly wildfire starts to remain at a moderate level. After transforming the forecasts back to the original fire-count scale, the point forecasts were mostly between about 7.5 and 10.4 wildfire starts per month. The prediction intervals were wide, showing that there is still a large amount of uncertainty when forecasting wildfire starts. Overall, the SARIMA model gives a useful baseline model for monthly wildfire starts, but wildfire activity is still difficult to predict exactly because it can depend on sudden weather, ignition, and environmental conditions.

4.2 ARMAX Results for Monthly Acres Burned

The second part of the analysis modeled monthly acres burned using an ARMAX model with external predictors. Because acres burned is highly right-skewed and has several extreme wildfire months, I used

$$z_t = \log(1 + \text{total acres burned}_t)$$

as the response variable.

The ARMAX analysis used the February 2000 through December 2020 sample. After merging wildfire data with drought, weather, Santa Ana wind, and NDVI variables, the final ARMAX dataset contained 248 complete monthly observations.

I tested a large set of possible predictors, including lagged SPI drought categories, temperature, wind speed, relative humidity, vapor pressure deficit, Santa Ana-like wind variables, and mean NDVI. BIC selection chose `D2_lag3`, `santa_ana_flag`, and `mean_ndvi` as the final predictors. The selected regression part of the model was approximately

$$\log(1 + \widehat{\text{acres burned}}) = 7.7355 - 0.0141(\text{D2_lag3}) + 2.2227(\text{santa_ana_flag}) - 12.5701(\text{mean_ndvi})$$

All three selected predictors had small p-values. The p-value for `D2_lag3` was about 0.0005, the p-value for `santa_ana_flag` was about 0.0023, and the p-value for `mean_ndvi` was about 3.85×10^{-8} . The model R^2 was about 0.175. This R^2 is not very high, but that is expected because acres burned is irregular and dominated by rare extreme wildfire months. The regression step was mainly used for variable selection before fitting the ARMAX model.

The VIF values for the selected predictors were all close to 1. This suggests that multicollinearity was not a major issue in the selected model.

After fitting the ARMAX model with the BIC-selected predictors, the model improved over the baseline acres-burned model. The baseline SARIMA model for transformed acres burned had AICc about 1094.54 and BIC about 1101.48. The ARMAX model with the selected predictors had AICc about 1065.62 and BIC about 1089.66. Since lower AICc and BIC values are better, this suggests that adding drought, Santa Ana-like wind conditions, and NDVI improved the model.

The final ARMAX model used regression with ARIMA(1,0,1) errors. The estimated ARMAX coefficients were similar to the regression screening model: lagged severe drought had a negative coefficient, the Santa Ana-like wind indicator had a positive coefficient, and mean NDVI had a negative coefficient. The Ljung-Box test for the ARMAX residuals had a p-value of about 0.8344, which is above 0.05. This means I do not reject the null hypothesis that the residuals are white noise. In other words, the ARMAX model captured most of the remaining autocorrelation in the transformed acres-burned series.

Overall, the ARMAX results suggest that monthly acres burned in Ventura County is related to lagged severe drought conditions, Santa Ana-like wind events, and vegetation greenness. The model does not perfectly explain wildfire severity, but it gives a stronger result than the baseline SARIMA model and connects well with known wildfire risk factors in Southern California.

5 Conclusion and Future Study

In this project, I studied monthly wildfire activity in Ventura County, California from February 2000 to December 2020 using time series methods. The first part of the project used a SARIMA model to analyze monthly wildfire starts. The selected SARIMA model was $ARIMA(0, 1, 3)(0, 0, 1)_{12}$, which used one regular difference, three nonseasonal moving-average terms, and one seasonal moving-average term. The Ljung-Box test for this model had a p-value above 0.05, so there was not strong evidence of remaining autocorrelation in the residuals. This suggests that the SARIMA model was a reasonable baseline model for monthly wildfire starts.

The second part of the project used an ARMAX model to study monthly acres burned. Since acres burned is more connected to wildfire severity, I added external predictors related to drought, weather, Santa Ana-like wind events, and vegetation conditions. After testing a large set of possible predictors, BIC selected lagged severe drought, a Santa Ana-like wind indicator, and mean NDVI as the final predictors. This result suggests that wildfire severity in Ventura County is connected to drought conditions, Southern California wind patterns, and vegetation greenness.

The final ARMAX model improved over the baseline acres-burned model based on both AICc and BIC. The baseline SARIMA model for acres burned had AICc about 1094.54 and BIC about 1101.48, while the ARMAX model had AICc about 1065.62 and BIC about 1089.66. The residual diagnostics were also acceptable because the Ljung-Box test did not show strong remaining autocorrelation. Overall, the ARMAX model gave a stronger explanation of monthly acres burned than a model based only on past values of the series.

There are several ways this project could be extended in the future. First, the Santa Ana wind variable used here is a simplified proxy based on wind direction, wind speed, and relative humidity. Future work could use an official Santa Ana wind event dataset or a more detailed regional wind classification. Second, the NDVI variable was based on vegetation greenness, but future work could include more direct fuel moisture measures if reliable station-level data were available. Third, a spatial analysis could be added by studying where fires occurred inside Ventura County instead of

only using county-level monthly totals. Finally, future models could test lagged wildfire counts as predictors of acres burned, since high fire activity in one month may indicate persistent fire-prone conditions in the following month.

Overall, this project shows that SARIMA and ARMAX models can be useful for studying wildfire time series. The SARIMA model gave a baseline model for wildfire starts, while the ARMAX model showed that drought, Santa Ana-like wind conditions, and vegetation conditions helped explain monthly acres burned in Ventura County.

6 References

NASA Earthdata. (2024). MODIS/Terra Vegetation Indices 16-Day L3 Global 250m SIN Grid, Version 6.1 (MOD13Q1). National Aeronautics and Space Administration.

NASA POWER Project. (2024). POWER Daily API: Prediction Of Worldwide Energy Resources. National Aeronautics and Space Administration.

NOAA National Integrated Drought Information System. (2024). U.S. Gridded Standardized Precipitation Index (SPI) from nClimGrid. Drought.gov.

Short, K. C. (2022). Spatial wildfire occurrence data for the United States, 1992-2020 [FPA_FOD_20221014], 6th Edition. Forest Service Research Data Archive.

7 Appendix

The appendix contains the main R code used for the final analysis. To make the SARIMA and ARMAX analyses more comparable, I restricted the working sample to February 2000 through December 2020. This matches the period used for the NDVI-based ARMAX model.

7.1 Appendix A: Libraries and Wildfire Data

```
library(readr)
library(dplyr)
library(lubridate)
library(ggplot2)
library(forecast)
library(MASS)
library(car)
library(jsonlite)
library(MODISTools)

# Define common study period
sample_start <- as.Date("2000-02-01")
sample_end <- as.Date("2020-12-01")

# Load monthly wildfire data
monthly_full <- read_csv("ventura_wildfires_monthly_1992_2020.csv",
                        show_col_types = FALSE) |>
  mutate(month = as.Date(month))

# Restrict to 2000-2020 sample
monthly_model <- monthly_full |>
  filter(
    month >= sample_start,
    month <= sample_end
  )

nrow(monthly_model)
```

```
[1] 251
```

```
summary(monthly_model)
```

month	fire_count	total_acres_burned
Min. :2000-02-01	Min. : 0.000	Min. : 0.00
1st Qu.:2005-04-16	1st Qu.: 0.000	1st Qu.: 0.00
Median :2010-07-01	Median : 1.000	Median : 0.10

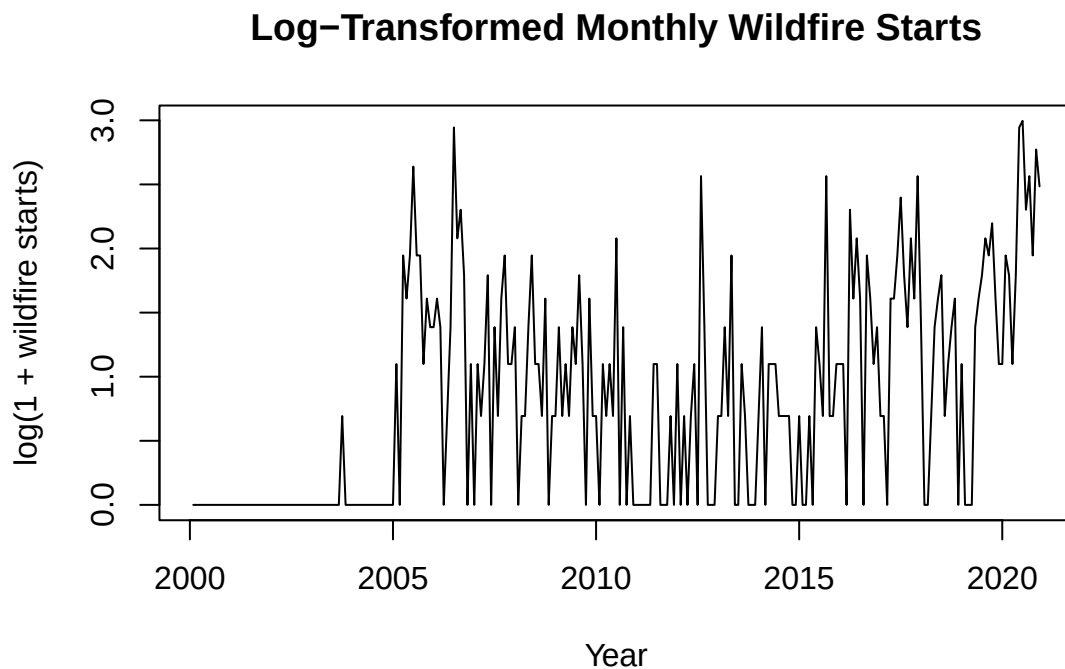
Mean	:2010-07-01	Mean	: 2.227	Mean	: 2446.68
3rd Qu.	:2015-09-16	3rd Qu.	: 3.000	3rd Qu.	: 3.75
Max.	:2020-12-01	Max.	:19.000	Max.	:281899.54

7.2 Appendix B: SARIMA Model for Monthly Wildfire Starts

```
# Create monthly fire-count time series using the common sample
fire_ts <- ts(
  monthly_model$fire_count,
  start = c(year(min(monthly_model$month)),
            month(min(monthly_model$month))),
  frequency = 12
)

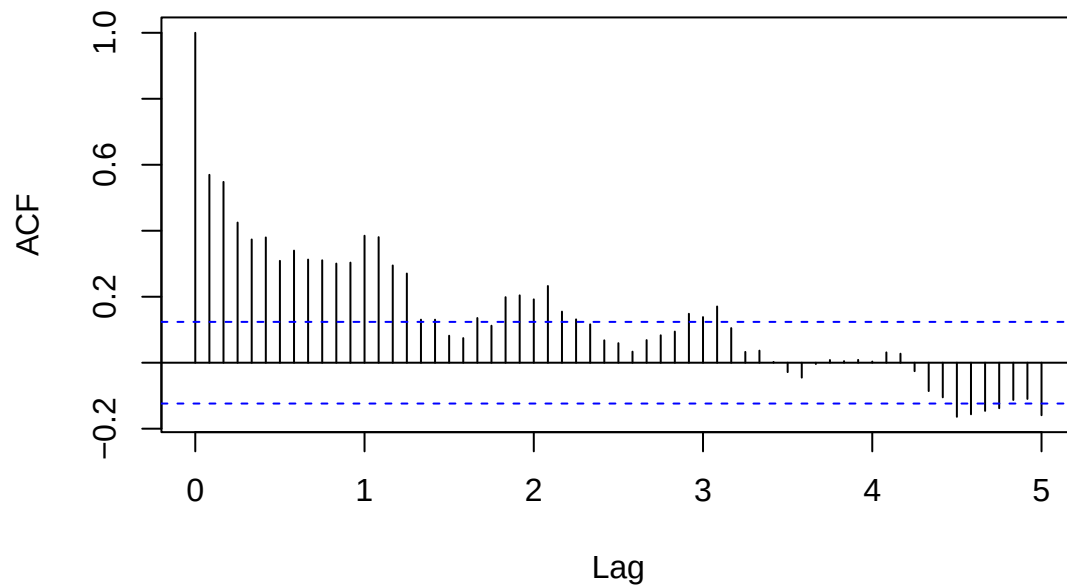
# Log transform fire count
fire_log <- log1p(fire_ts)

plot(fire_log,
     main = "Log-Transformed Monthly Wildfire Starts",
     ylab = "log(1 + wildfire starts)",
     xlab = "Year")
```



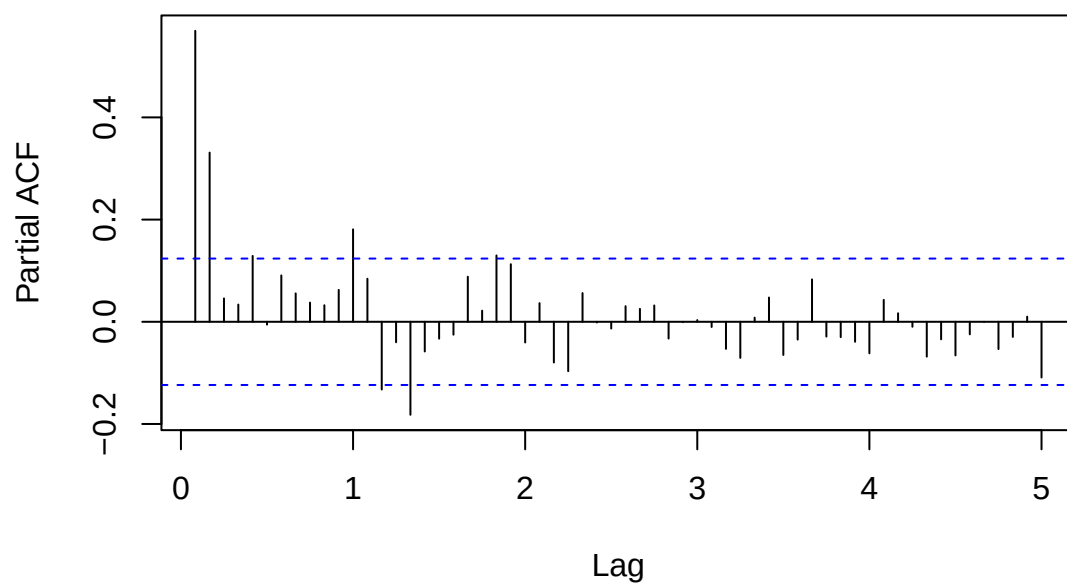
```
acf(fire_log,
    lag.max = 60,
    main = "ACF of Log-Transformed Wildfire Starts")
```

ACF of Log-Transformed Wildfire Starts



```
pacf(fire_log,  
     lag.max = 60,  
     main = "PACF of Log-Transformed Wildfire Starts")
```

PACF of Log-Transformed Wildfire Starts



```
# Check differencing
ndiffs(fire_log)
```

```
[1] 1
```

```
nsdiffs(fire_log)
```

```
[1] 0
```

```
# Fit SARIMA model
fit_auto <- auto.arima(
  fire_log,
  seasonal = TRUE,
  stepwise = FALSE,
  approximation = FALSE
)

summary(fit_auto)
```

```
Series: fire_log
ARIMA(0,1,3)(0,0,1)[12]
```

```
Coefficients:
```

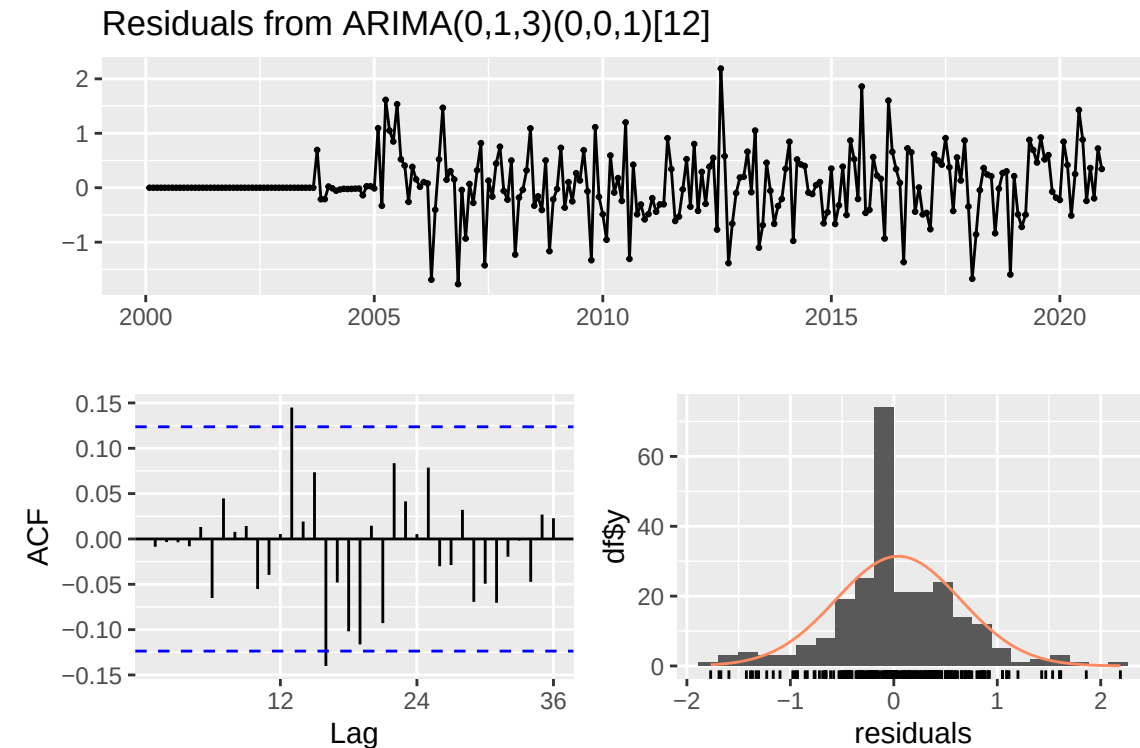
	ma1	ma2	ma3	sma1
	-0.6952	0.0907	-0.2149	0.1811
s.e.	0.0624	0.0740	0.0624	0.0628

```
sigma^2 = 0.3692: log likelihood = -228.84
AIC=467.67 AICc=467.92 BIC=485.28
```

```
Training set error measures:
```

	ME	RMSE	MAE	MPE	MAPE	MASE	ACF1
Training set	0.04285571	0.6015536	0.4171235	NaN	Inf	0.7122201	-0.008611779

```
checkresiduals(fit_auto)
```



Ljung-Box test

```
data: Residuals from ARIMA(0,1,3)(0,0,1)[12]
Q* = 27.474, df = 20, p-value = 0.1224
```

Model df: 4. Total lags used: 24

```
# Forecast 12 months ahead
wildfire_forecast <- forecast(fit_auto, h = 12)
```

```
forecast_counts <- data.frame(
  month_ahead = 1:12,
  point_forecast = expm1(wildfire_forecast$mean),
  lower_80 = expm1(wildfire_forecast$lower[, 1]),
  upper_80 = expm1(wildfire_forecast$upper[, 1]),
  lower_95 = expm1(wildfire_forecast$lower[, 2]),
  upper_95 = expm1(wildfire_forecast$upper[, 2])
)
```

```
forecast_counts
```

```
  month_ahead point_forecast lower_80 upper_80 lower_95 upper_95
1           1           9.081959  3.627563 20.96532  2.064249 32.17155
```

2	2	9.677169	3.730485	23.09942	2.074308	36.08215
3	3	8.647800	3.040413	22.03726	1.548727	35.52021
4	4	7.533581	2.533690	19.60792	1.215806	31.86478
5	5	8.282030	2.801033	21.66649	1.369429	35.36153
6	6	10.364974	3.603077	27.06006	1.852730	44.27685
7	7	10.408493	3.570743	27.47540	1.816436	45.21221
8	8	8.906665	2.926631	23.99394	1.405821	39.79357
9	9	9.467387	3.105052	25.69057	1.501037	42.80830
10	10	8.289345	2.604997	22.93676	1.184199	38.50736
11	11	10.018765	3.231966	27.68954	1.550009	46.61284
12	12	9.524487	3.000812	26.68559	1.397649	45.19727

7.3 Appendix C: External Predictors for ARMAX

```
# Appendix C: External Predictors for ARMAX
# This chunk creates SPI drought variables, NASA POWER weather variables,
# a Santa Ana-like wind proxy, VPD, and MODIS NDVI.

# 1. Define common sample and make sure wildfire data exists

sample_start <- as.Date("2000-02-01")
sample_end <- as.Date("2020-12-01")

if (!exists("monthly_full")) {
  monthly_full <- readr::read_csv("ventura_wildfires_monthly_1992_2020.csv",
                                  show_col_types = FALSE) |>
    dplyr::mutate(month = as.Date(month))
}

monthly_model <- monthly_full |>
  dplyr::filter(
    month >= sample_start,
    month <= sample_end
  )

# 2. Load and clean SPI drought data

spi_raw <- readr::read_csv("SPI-06111.csv",
                           show_col_types = FALSE)

spi_clean <- spi_raw |>
  dplyr::mutate(
    month = as.Date(gsub("d_", "", DATE), format = "%Y%m%d")
  ) |>
  dplyr::filter(
    month >= sample_start,
    month <= sample_end
  ) |>
  dplyr::select(month, D0, D1, D2, D3, D4)

# 3. Merge wildfire and SPI data, then create drought lags

wildfire_spi <- monthly_model |>
  dplyr::left_join(spi_clean, by = "month") |>
  dplyr::arrange(month) |>
  dplyr::mutate(
    log_acres = log1p(total_acres_burned),
    month_num = lubridate::month(month),
```

```

weighted_dry_index = D0 + 2*D1 + 3*D2 + 4*D3 + 5*D4,
weighted_dry_lag1 = dplyr::lag(weighted_dry_index, 1),
weighted_dry_lag3 = dplyr::lag(weighted_dry_index, 3),
weighted_dry_lag6 = dplyr::lag(weighted_dry_index, 6),

D0_lag1 = dplyr::lag(D0, 1),
D1_lag1 = dplyr::lag(D1, 1),
D2_lag1 = dplyr::lag(D2, 1),
D3_lag1 = dplyr::lag(D3, 1),
D4_lag1 = dplyr::lag(D4, 1),

D0_lag3 = dplyr::lag(D0, 3),
D1_lag3 = dplyr::lag(D1, 3),
D2_lag3 = dplyr::lag(D2, 3),
D3_lag3 = dplyr::lag(D3, 3),
D4_lag3 = dplyr::lag(D4, 3),

D0_lag6 = dplyr::lag(D0, 6),
D1_lag6 = dplyr::lag(D1, 6),
D2_lag6 = dplyr::lag(D2, 6),
D3_lag6 = dplyr::lag(D3, 6),
D4_lag6 = dplyr::lag(D4, 6)
)

# 4. Download or load NASA POWER weather data

lat <- 34.37
lon <- -119.14

weather_file <- "ventura_weather_monthly_2000_2020.csv"

if (file.exists(weather_file)) {

  weather_monthly_vpd <- readr::read_csv(weather_file,
                                         show_col_types = FALSE) |>
    dplyr::mutate(month = as.Date(month))

} else {

  weather_start <- format(sample_start, "%Y%m%d")
  weather_end <- format(lubridate::ceiling_date(sample_end, "month") - lubridate::days(1),
                       "%Y%m%d")

  power_url_vpd <- paste0(
    "https://power.larc.nasa.gov/api/temporal/daily/point?",
    "parameters=T2M,T2M_MAX,RH2M,WS10M,WS10M_MAX,WD10M",
    "&community=AG",

```

```

"&longitude=", lon,
"&latitude=", lat,
"&start=", weather_start,
"&end=", weather_end,
"&format=JSON"
)

power_raw_vpd <- jsonlite::fromJSON(power_url_vpd)

params <- power_raw_vpd$properties$parameter

weather_daily_vpd <- data.frame(
  date = as.Date(names(params$T2M), format = "%Y%m%d"),
  tmean = as.numeric(params$T2M),
  tmax = as.numeric(params$T2M_MAX),
  rh = as.numeric(params$RH2M),
  wind_mean = as.numeric(params$WS10M),
  wind_max = as.numeric(params$WS10M_MAX),
  wind_dir = as.numeric(params$WD10M)
) |>
dplyr::mutate(
  es = 0.6108 * exp((17.27 * tmean) / (tmean + 237.3)),
  vpd = es * (1 - rh / 100),

  santa_ana_day = dplyr::if_else(
    !is.na(wind_dir) & !is.na(wind_max) & !is.na(rh) &
    wind_dir >= 20 & wind_dir <= 120 &
    wind_max >= 6 & rh <= 30,
    1, 0
  )
)

weather_monthly_vpd <- weather_daily_vpd |>
dplyr::mutate(month = lubridate::floor_date(date, "month")) |>
dplyr::group_by(month) |>
dplyr::summarize(
  mean_tmax = mean(tmax, na.rm = TRUE),
  mean_rh = mean(rh, na.rm = TRUE),
  mean_vpd = mean(vpd, na.rm = TRUE),
  max_vpd = max(vpd, na.rm = TRUE),
  mean_wind = mean(wind_mean, na.rm = TRUE),
  max_wind = max(wind_max, na.rm = TRUE),
  santa_ana_days = sum(santa_ana_day, na.rm = TRUE),
  santa_ana_flag = as.integer(sum(santa_ana_day, na.rm = TRUE) > 0),
  .groups = "drop"
)

```

```
  readr::write_csv(weather_monthly_vpd, weather_file)
}

# 5. Download or load MODIS NDVI data

ndvi_file <- "ventura_ndvi_monthly_2000_2020.csv"

if (file.exists(ndvi_file)) {

  ndvi_monthly <- readr::read_csv(ndvi_file,
                                show_col_types = FALSE) |>
    dplyr::mutate(month = as.Date(month))

} else {

  ndvi_raw <- MODISTools::mt_subset(
    product = "MOD13Q1",
    lat = lat,
    lon = lon,
    band = "250m_16_days_NDVI",
    start = "2000-02-18",
    end = "2020-12-31",
    km_lr = 20,
    km_ab = 20,
    site_name = "Ventura_County"
  )

  ndvi_monthly <- ndvi_raw |>
    dplyr::mutate(
      date = as.Date(calendar_date),
      ndvi = value * 0.0001,
      month = lubridate::floor_date(date, "month")
    ) |>
    dplyr::filter(
      !is.na(ndvi),
      ndvi >= -1,
      ndvi <= 1
    ) |>
    dplyr::group_by(month) |>
    dplyr::summarize(
      mean_ndvi = mean(ndvi, na.rm = TRUE),
      .groups = "drop"
    )

  readr::write_csv(ndvi_monthly, ndvi_file)
}
```

```
# 6. Merge wildfire, drought, weather, and NDVI data
model_data_vpd_ndvi <- wildfire_spi |>
  dplyr::left_join(weather_monthly_vpd, by = "month") |>
  dplyr::left_join(ndvi_monthly, by = "month") |>
  dplyr::filter(
    month >= sample_start,
    month <= sample_end,
    !is.na(log_acres),
    !is.na(D2_lag3),
    !is.na(mean_ndvi),
    !is.na(santa_ana_flag)
  )

nrow(model_data_vpd_ndvi)
```

```
[1] 248
```

```
summary(model_data_vpd_ndvi)
```

month	fire_count	total_acres_burned	D0
Min. :2000-05-01	Min. : 0.000	Min. :0.000e+00	Min. : 0.00
1st Qu.:2005-06-23	1st Qu.: 0.000	1st Qu.:0.000e+00	1st Qu.: 0.00
Median :2010-08-16	Median : 1.000	Median :1.100e-01	Median : 13.85
Mean :2010-08-16	Mean : 2.254	Mean :2.476e+03	Mean : 43.94
3rd Qu.:2015-10-08	3rd Qu.: 3.000	3rd Qu.:4.003e+00	3rd Qu.:100.00
Max. :2020-12-01	Max. :19.000	Max. :2.819e+05	Max. :100.00

D1	D2	D3	D4
Min. : 0.00	Min. : 0.00	Min. : 0.00	Min. : 0.00
1st Qu.: 0.00	1st Qu.: 0.00	1st Qu.: 0.00	1st Qu.: 0.00
Median : 0.00	Median : 0.00	Median : 0.00	Median : 0.00
Mean : 33.89	Mean : 18.68	Mean : 12.26	Mean : 4.51
3rd Qu.: 85.88	3rd Qu.: 10.07	3rd Qu.: 0.00	3rd Qu.: 0.00
Max. :100.00	Max. :100.00	Max. :100.00	Max. :100.00

log_acres	month_num	weighted_dry_index	weighted_dry_lag1
Min. : 0.0000	Min. : 1.000	Min. : 0.00	Min. : 0.00
1st Qu.: 0.0000	1st Qu.: 4.000	1st Qu.: 0.00	1st Qu.: 0.00
Median : 0.1044	Median : 7.000	Median : 17.35	Median : 17.35
Mean : 1.2960	Mean : 6.565	Mean : 239.35	Mean : 239.35
3rd Qu.: 1.6099	3rd Qu.:10.000	3rd Qu.: 295.05	3rd Qu.: 295.05
Max. :12.5493	Max. :12.000	Max. :1500.00	Max. :1500.00

weighted_dry_lag3	weighted_dry_lag6	D0_lag1	D1_lag1
Min. : 0.00	Min. : 0.0	Min. : 0.00	Min. : 0.00
1st Qu.: 0.00	1st Qu.: 0.0	1st Qu.: 0.00	1st Qu.: 0.00

Median : 17.35	Median : 18.4	Median : 13.85	Median : 0.00
Mean : 239.41	Mean : 242.2	Mean : 43.94	Mean : 33.89
3rd Qu.: 295.05	3rd Qu.: 307.2	3rd Qu.:100.00	3rd Qu.: 85.88
Max. :1500.00	Max. :1500.0	Max. :100.00	Max. :100.00

NA's :3

D2_lag1	D3_lag1	D4_lag1	D0_lag3
Min. : 0.00	Min. : 0.00	Min. : 0.00	Min. : 0.00
1st Qu.: 0.00	1st Qu.: 0.00	1st Qu.: 0.00	1st Qu.: 0.00
Median : 0.00	Median : 0.00	Median : 0.00	Median : 13.85
Mean : 18.68	Mean : 12.26	Mean : 4.51	Mean : 44.00
3rd Qu.: 10.07	3rd Qu.: 0.00	3rd Qu.: 0.00	3rd Qu.:100.00
Max. :100.00	Max. :100.00	Max. :100.00	Max. :100.00

D1_lag3	D2_lag3	D3_lag3	D4_lag3
Min. : 0.00	Min. : 0.00	Min. : 0.00	Min. : 0.00
1st Qu.: 0.00	1st Qu.: 0.00	1st Qu.: 0.00	1st Qu.: 0.00
Median : 0.00	Median : 0.00	Median : 0.00	Median : 0.00
Mean : 33.89	Mean : 18.68	Mean : 12.26	Mean : 4.51
3rd Qu.: 85.88	3rd Qu.: 10.07	3rd Qu.: 0.00	3rd Qu.: 0.00
Max. :100.00	Max. :100.00	Max. :100.00	Max. :100.00

D0_lag6	D1_lag6	D2_lag6	D3_lag6
Min. : 0.00	Min. : 0.0	Min. : 0.0	Min. : 0.00
1st Qu.: 0.00	1st Qu.: 0.0	1st Qu.: 0.0	1st Qu.: 0.00
Median : 14.90	Median : 0.0	Median : 0.0	Median : 0.00
Mean : 44.38	Mean : 34.3	Mean : 18.9	Mean : 12.41
3rd Qu.:100.00	3rd Qu.: 86.4	3rd Qu.: 11.8	3rd Qu.: 0.00
Max. :100.00	Max. :100.0	Max. :100.0	Max. :100.00
NA's :3	NA's :3	NA's :3	NA's :3

D4_lag6	mean_tmax	mean_rh	mean_vpd
Min. : 0.000	Min. :10.54	Min. :28.93	Min. :0.1610
1st Qu.: 0.000	1st Qu.:17.98	1st Qu.:40.52	1st Qu.:0.5227
Median : 0.000	Median :22.94	Median :52.19	Median :0.8689
Mean : 4.566	Mean :23.93	Mean :52.54	Mean :1.0749
3rd Qu.: 0.000	3rd Qu.:30.91	3rd Qu.:62.16	3rd Qu.:1.6635
Max. :100.000	Max. :35.84	Max. :83.46	Max. :2.4659
NA's :3			

max_vpd	mean_wind	max_wind	santa_ana_days
Min. :0.4321	Min. :1.760	Min. : 3.730	Min. :0.00000
1st Qu.:1.0953	1st Qu.:2.188	1st Qu.: 4.845	1st Qu.:0.00000
Median :1.7870	Median :2.484	Median : 5.825	Median :0.00000
Mean :1.9389	Mean :2.467	Mean : 5.936	Mean :0.04435
3rd Qu.:2.8087	3rd Qu.:2.720	3rd Qu.: 6.915	3rd Qu.:0.00000
Max. :4.2814	Max. :3.471	Max. :10.310	Max. :2.00000

santa_ana_flag	mean_ndvi
Min. :0.00000	Min. :0.3531
1st Qu.:0.00000	1st Qu.:0.4466

Median	:0.00000	Median	:0.4903
Mean	:0.03629	Mean	:0.4966
3rd Qu.	:0.00000	3rd Qu.	:0.5468
Max.	:1.00000	Max.	:0.6349

7.4 Appendix D: BIC Selection and Final ARMAX Model

```
# Candidate predictors for ARMAX selection
candidate_predictors <- c(
  "D0_lag1", "D1_lag1", "D2_lag1", "D3_lag1", "D4_lag1",
  "D0_lag3", "D1_lag3", "D2_lag3", "D3_lag3", "D4_lag3",
  "D0_lag6", "D1_lag6", "D2_lag6", "D3_lag6", "D4_lag6",

  "mean_tmax", "max_tmax",
  "mean_wind", "max_wind",
  "mean_rh",
  "mean_vpd", "max_vpd",

  "santa_ana_days", "santa_ana_flag",

  "mean_ndvi"
)

candidate_predictors <- candidate_predictors[
  candidate_predictors %in% names(model_data_vpd_ndvi)
]

model_data_big <- model_data_vpd_ndvi |>
  dplyr::select(month, log_acres, all_of(candidate_predictors)) |>
  na.omit()

big_formula <- reformulate(
  termlabels = candidate_predictors,
  response = "log_acres"
)

big_lm <- lm(
  big_formula,
  data = model_data_big
)

# BIC stepwise selection
big_bic_lm <- stepAIC(
  big_lm,
  direction = "both",
  k = log(nrow(model_data_big)),
  trace = FALSE
)

summary(big_bic_lm)
```


Call:

```
lm(formula = log_acres ~ D2_lag3 + santa_ana_flag + mean_ndvi,
    data = model_data_big)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-3.8177	-1.2261	-0.5701	0.7086	10.3081

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	7.735505	1.138064	6.797	8.32e-11	***
D2_lag3	-0.014069	0.004004	-3.514	0.000528	***
santa_ana_flag	2.222711	0.721931	3.079	0.002319	**
mean_ndvi	-12.570137	2.212847	-5.681	3.85e-08	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.103 on 241 degrees of freedom

Multiple R-squared: 0.175, Adjusted R-squared: 0.1647

F-statistic: 17.04 on 3 and 241 DF, p-value: 4.536e-10

```
selected_big_terms <- attr(terms(big_bic_lm), "term.labels")
selected_big_terms
```

```
[1] "D2_lag3"          "santa_ana_flag" "mean_ndvi"
```

```
# VIF
num_predictors <- length(coef(big_bic_lm)) - 1

if (num_predictors >= 2) {
  vif(big_bic_lm)
} else {
  cat("VIF not calculated because the selected model has only one predictor.\n")
}
```

D2_lag3	santa_ana_flag	mean_ndvi
1.126690	1.022096	1.144383

```
#Create acres-burned timeseries
log_acres_big_ts <- ts(
  model_data_big$log_acres,
  start = c(year(min(model_data_big$month)),
            month(min(model_data_big$month))),
  frequency = 12
)
```

```
#SARIMA for acres burned
baseline_big <- auto.arima(
  log_acres_big_ts,
  seasonal = TRUE,
  stepwise = FALSE,
  approximation = FALSE
)

#ARMAX with BIC-selected predictors
xreg_big_selected <- model_data_big |>
  dplyr::select(all_of(selected_big_terms)) |>
  as.matrix()

armax_big_selected <- auto.arima(
  log_acres_big_ts,
  xreg = xreg_big_selected,
  seasonal = TRUE,
  stepwise = FALSE,
  approximation = FALSE
)

summary(baseline_big)
```

Series: log_acres_big_ts
ARIMA(0,1,1)

Coefficients:

```
      ma1
    -0.9354
s.e.    0.0242
```

```
sigma^2 = 5.088:  log likelihood = -545.24
AIC=1094.49   AICc=1094.54   BIC=1101.48
```

Training set error measures:

	ME	RMSE	MAE	MPE	MAPE	MASE	ACF1
Training set	0.1686432	2.246469	1.390099	-Inf	Inf	0.7565027	0.01863098

```
summary(armax_big_selected)
```

Series: log_acres_big_ts
Regression with ARIMA(1,0,1) errors

Coefficients:

```
      ar1      ma1  intercept  D2_lag3  santa_ana_flag  mean_ndvi
0.9309  -0.8784      7.5268  -0.0126           2.1821  -12.2405
```

```
s.e. 0.0736 0.0904 1.1948 0.0044 0.7069 2.3038
```

```
sigma^2 = 4.38: log likelihood = -525.57
```

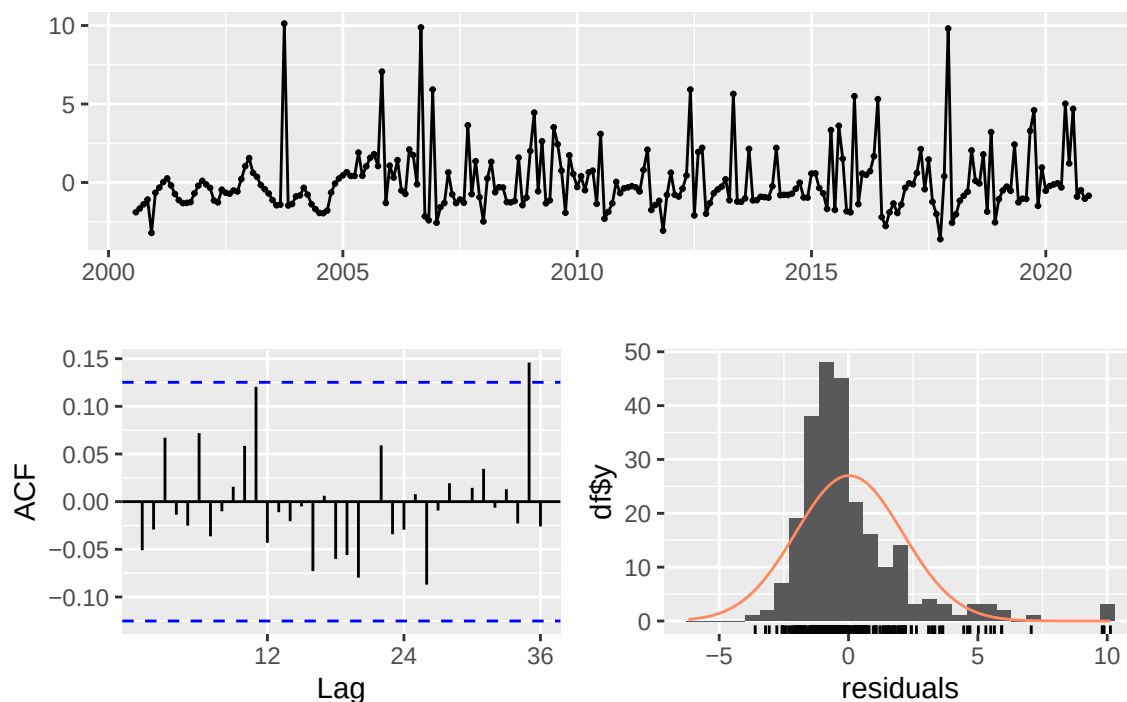
```
AIC=1065.15 AICc=1065.62 BIC=1089.66
```

Training set error measures:

	ME	RMSE	MAE	MPE	MAPE	MASE	ACF1
Training set	0.02482979	2.066966	1.424296	NaN	Inf	0.775113	-0.05094869

```
checkresiduals(armax_big_selected)
```

Residuals from Regression with ARIMA(1,0,1) errors



Ljung-Box test

```
data: Residuals from Regression with ARIMA(1,0,1) errors
```

```
Q* = 15.618, df = 22, p-value = 0.8344
```

```
Model df: 2. Total lags used: 24
```

```
big_model_comparison <- data.frame(
  model = c(
    "Baseline SARIMA, same sample",
    "ARMAX with BIC-selected predictors"
  ),

```

```

AICc = c(
  baseline_big$aicc,
  armax_big_selected$aicc
),
BIC = c(
  BIC(baseline_big),
  BIC(armax_big_selected)
)
)

big_model_comparison

```

	model	AICc	BIC
1	Baseline SARIMA, same sample	1094.537	1101.481
2	ARMAX with BIC-selected predictors	1065.619	1089.656

```
coef(big_bic_lm)
```

(Intercept)	D2_lag3	santa_ana_flag	mean_ndvi
7.73550550	-0.01406856	2.22271123	-12.57013722

```
selected_big_terms
```

```
[1] "D2_lag3"      "santa_ana_flag" "mean_ndvi"
```